

Lesson 12: Text, Informational and Logical Functions in Power BI

Prerequisites: Connect SQL Server to Power BI desktop and import DimCustomer table from AdventureWorksDW database (2019)

Basic Level (1–10)

1. Concatenate First and Last Name

```
FullName = DimCustomer[FirstName] & " " & DimCustomer[LastName]
```

2. Convert Email Address to Uppercase

```
Email_Upper = UPPER(DimCustomer[EmailAddress])
```

3. Extract First 3 Characters from First Name

```
First3Chars = LEFT(DimCustomer[FirstName], 3)
```

4. Count Characters in Last Name

```
LastName_Length = LEN(DimCustomer[LastName])
```

5. Convert First Name to Lowercase

```
FirstName_Lower = LOWER(DimCustomer[FirstName])
```

6. Trim Spaces in EnglishEducation

```
Education_Trim = TRIM(DimCustomer[EnglishEducation])
```

7. Repeat '*' Equal to Length of First Name

```
StarMask = REPT(" ", LEN(DimCustomer[FirstName]))
```

8. Get Last 4 Characters of Phone Number

```
Phone_Last4 = RIGHT(DimCustomer[Phone], 4)
```

9. Format YearlyIncome to Currency with 2 Decimals

```
Income_Format = FORMAT(DimCustomer[YearlyIncome], "$#,##0.00")
```

10. Check If FirstName and LastName Are Exactly the Same

```
IsSame = IF(DimCustomer[FirstName] = DimCustomer[LastName], "Yes", "No")
```

Intermediate Level (11–20)

11. Find If 'Manager' Appears in Occupation (Case Sensitive)

```
HasManager = IF(SEARCH("Manager", DimCustomer[EnglishOccupation],,0) > 0, "Yes", "No")
```

12. Search for 'graduate' in EnglishEducation (Case Insensitive)

```
HasGraduate = IF(SEARCH("graduate", LOWER(DimCustomer[EnglishEducation]),,0) > 0, "Yes", "No")
```

13. Extract Characters 3–7 from First Name

```
FirstName_3_7 = MID(DimCustomer[FirstName], 3, 5)
```

14. Replace Area Code in Phone Number with 'XXX'

```
Phone_Masked = "XXX" & MID(DimCustomer[Phone], 4, LEN(DimCustomer[Phone]) - 3)
```

15. Format BirthDate as 'DD-MM-YYYY'

```
BirthDate_Format = FORMAT(DimCustomer[BirthDate], "DD-MM-YYYY")
```

16. Create Initial + Last Name Format

```
Initial_Last = LEFT(DimCustomer[FirstName],1) & "." &
DimCustomer[LastName]
```

17. Capitalize First Letter of FirstName

```
FirstName_Proper = UPPER(LEFT(DimCustomer[FirstName],1)) &
LOWER(MID(DimCustomer[FirstName],2,LEN(DimCustomer[FirstName])))
```

18. Substitute Dashes with Spaces in Phone

```
Phone_Spaces = SUBSTITUTE(DimCustomer[Phone], "-", " ")
```

19. Convert BirthDate Year to Numeric

```
BirthYear = VALUE(FORMAT(DimCustomer[BirthDate], "YYYY"))
```

20. Show YearlyIncome Rounded to 1 Decimal Without Commas

```
Income_Round = FORMAT(ROUND(DimCustomer[YearlyIncome],1), "0.0")
```

Advanced Level (21–30)**21. Customer Code: First 2 Letters of LastName + Last 2 of CustomerKey**

```
CustomerCode = LEFT(DimCustomer[LastName],2) &
RIGHT(DimCustomer[CustomerKey],2)
```

22. Validate Email Ends with '.com' and Contains '@'

```
ValidEmail = IF(AND(RIGHT(DimCustomer[EmailAddress],4) = ".com",
CONTAINSSTRING(DimCustomer[EmailAddress],"@")), "Valid","Invalid")
```

23. Extract Domain Name from EmailAddress

```
EmailDomain = RIGHT(DimCustomer[EmailAddress],
LEN(DimCustomer[EmailAddress]) - FIND("@", DimCustomer[EmailAddress]))
```

24. Mask Phone Number Except Last 4 Digits

```
PhoneMask = REPT("*", LEN(DimCustomer[Phone])-4) &
RIGHT(DimCustomer[Phone],4)
```

25. Proper Casing of Last Name (manual)

```
LastName_Proper = UPPER(LEFT(DimCustomer[LastName],1)) &
LOWER(MID(DimCustomer[LastName],2,LEN(DimCustomer[LastName])))
```

26. Replace Multiple Spaces in Occupation with Single Space

```
Occupation_Clean = SUBSTITUTE(DimCustomer[EnglishOccupation], " ", " ")
```

27. Generate Custom ID: Initials + Birth Year

```
CustomID = LEFT(DimCustomer[FirstName],1) & LEFT(DimCustomer[LastName],1)
& "_" & FORMAT(DimCustomer[BirthDate],"YYYY")
```

28. Remove Hyphens and Convert Phone to Number

```
Phone_Number = VALUE(SUBSTITUTE(DimCustomer[Phone], "-", ""))
```

29. Categorize Customers into Segments

```
Segment = SWITCH(TRUE(), DimCustomer[EnglishEducation]="Graduate Degree"
&& DimCustomer[YearlyIncome]>90000, "Elite",
DimCustomer[EnglishEducation]="Bachelors" &&
DimCustomer[YearlyIncome]>=60000 && DimCustomer[YearlyIncome]<=90000,
"Professional", DimCustomer[EnglishEducation]="High School", "Basic",
"Other")
```

30. Measure for Customer Count by Gender

```
CustomerCount = VAR SelGender = VALUES(DimCustomer[Gender]) RETURN
IF(COUNTROWS(SelGender)=0, COUNTROWS(DimCustomer),
```

```
IF(COUNTROWS(SelGender)=1, COUNTROWS(DimCustomer), "Multiple Values  
Selected"))
```